MODULE 3 · LECTURE 14-a

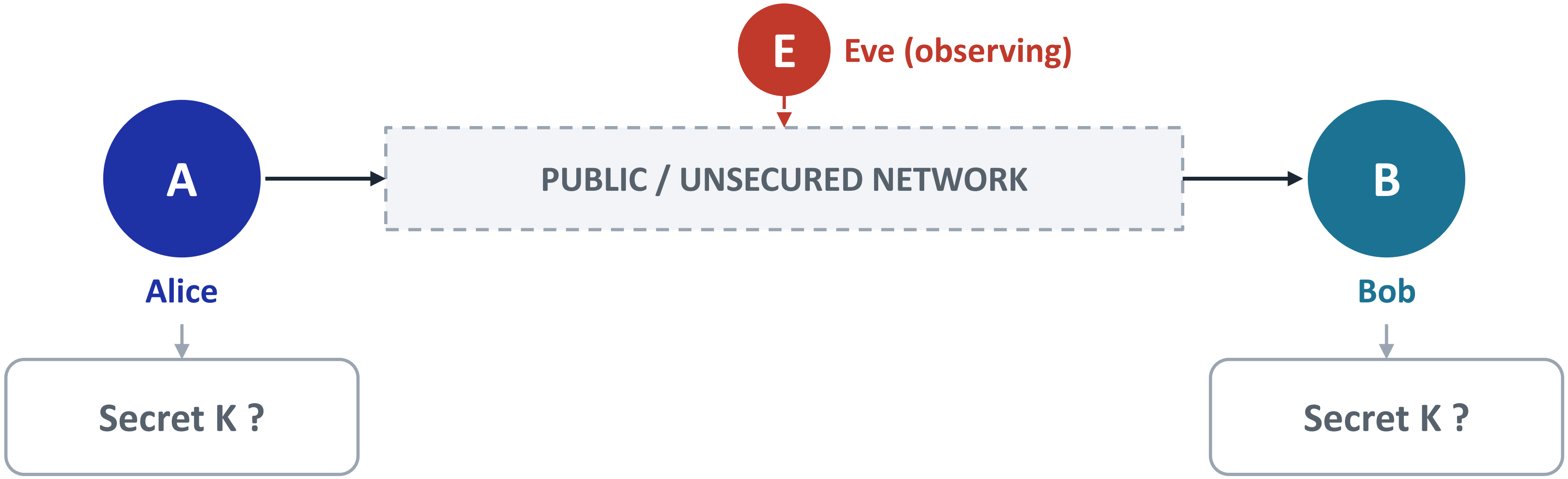
CS 6530 Applied Cryptography

Diffie–Hellman Key Agreement

Module 3 – Public-Key Cryptography

DIFFIE–HELLMAN KEY AGREEMENT

How Can Alice and Bob Establish a Shared Secret?


E
Eve (observing)
A
Alice
PUBLIC / UNSECURED NETWORK
B
Bob
Secret K ?
Secret K ?

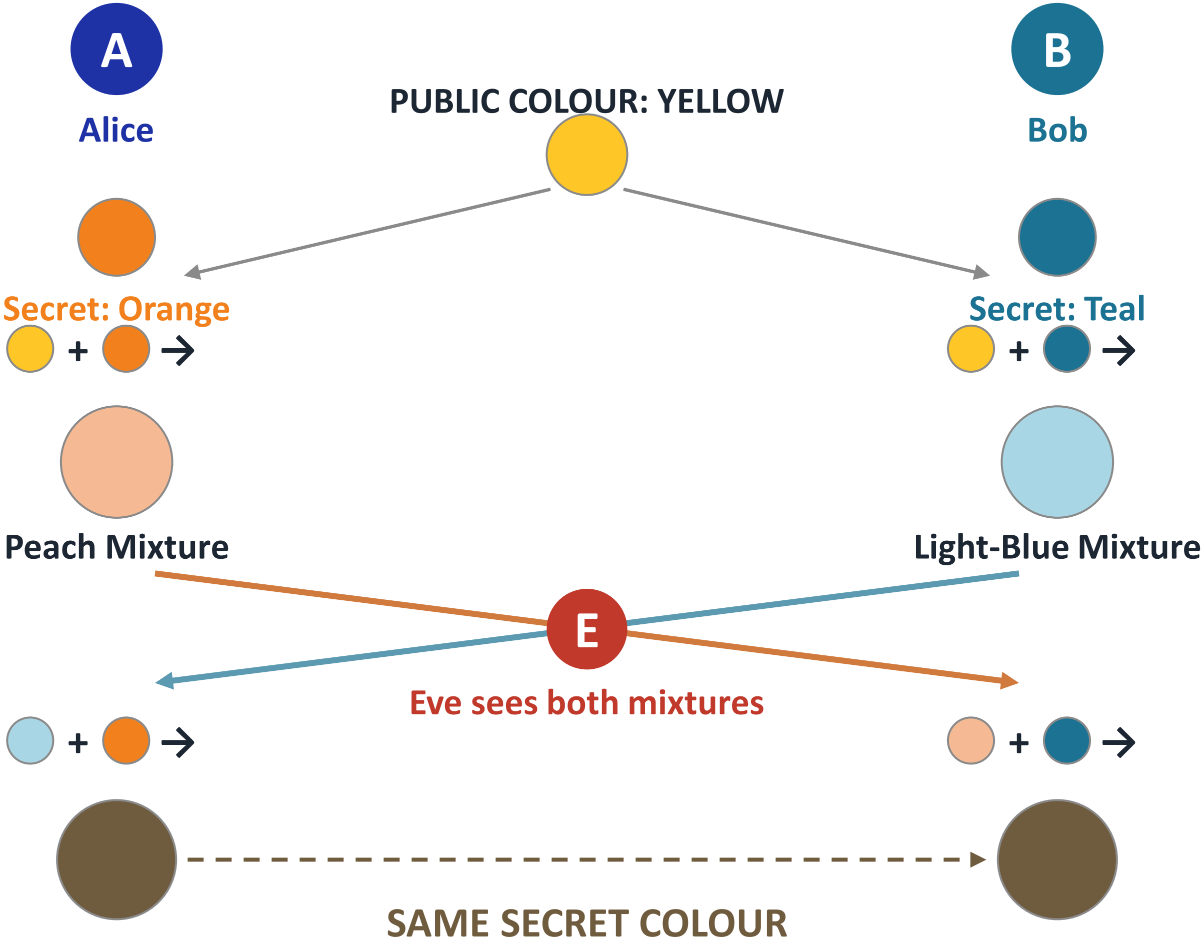
The Challenge

- No pre-shared secret
- Eve can observe the channel
- Alice and Bob need the same secret

Can Alice and Bob establish a shared secret without ever sending that secret?

INTUITION

Easy to Mix — Hard to Unmix



A

Alice

PUBLIC COLOUR: YELLOW

B

Bob

Secret: Orange

Secret: Teal

Peach Mixture

Light-Blue Mixture

E

Eve sees both mixtures

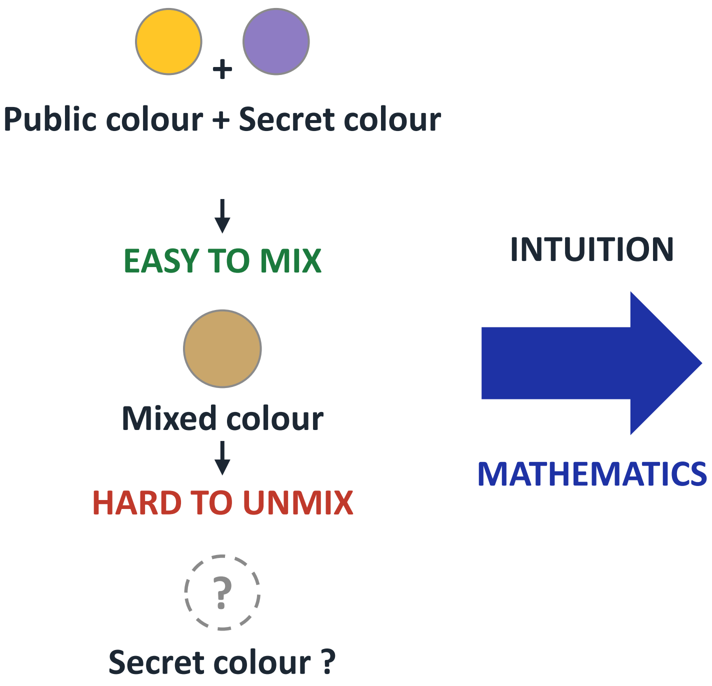
SAME SECRET COLOUR

An intuition only — next, we replace colours with mathematics.

Easy to mix. Hard to unmix.

FROM INTUITION TO MATHEMATICS

An Easy Forward Operation — A Hard Reverse Problem



Public colour + Secret colour

EASY TO MIX

Mixed colour

HARD TO UNMIX

Secret colour ?

INTUITION

MATHEMATICS

$$Y = g^x \bmod p$$

EASY

Given: g, x, p

Compute $Y = g^x \bmod p$

Modular exponentiation

HARD

Given: g, Y, p

Find x: $Y = g^x \bmod p$

Discrete logarithm problem

Public (p, g) + **Private x** → **Public $Y = g^x \bmod p$**

Modular exponentiation is easy; reversing it through the discrete logarithm problem is computationally hard for suitable parameters.

DIFFIE–HELLMAN SETUP

Public Parameters, Private Choices

PUBLIC PARAMETERS

p = large prime, g = suitable generator

A

Alice

Chooses randomly:

a

PRIVATE

↓

Computes:

$$A = g^a \bmod p$$

PUBLIC

B

Bob

Chooses randomly:

b

PRIVATE

↓

Computes:

$$B = g^b \bmod p$$

PUBLIC

E

Eve can know: p, g, A, B

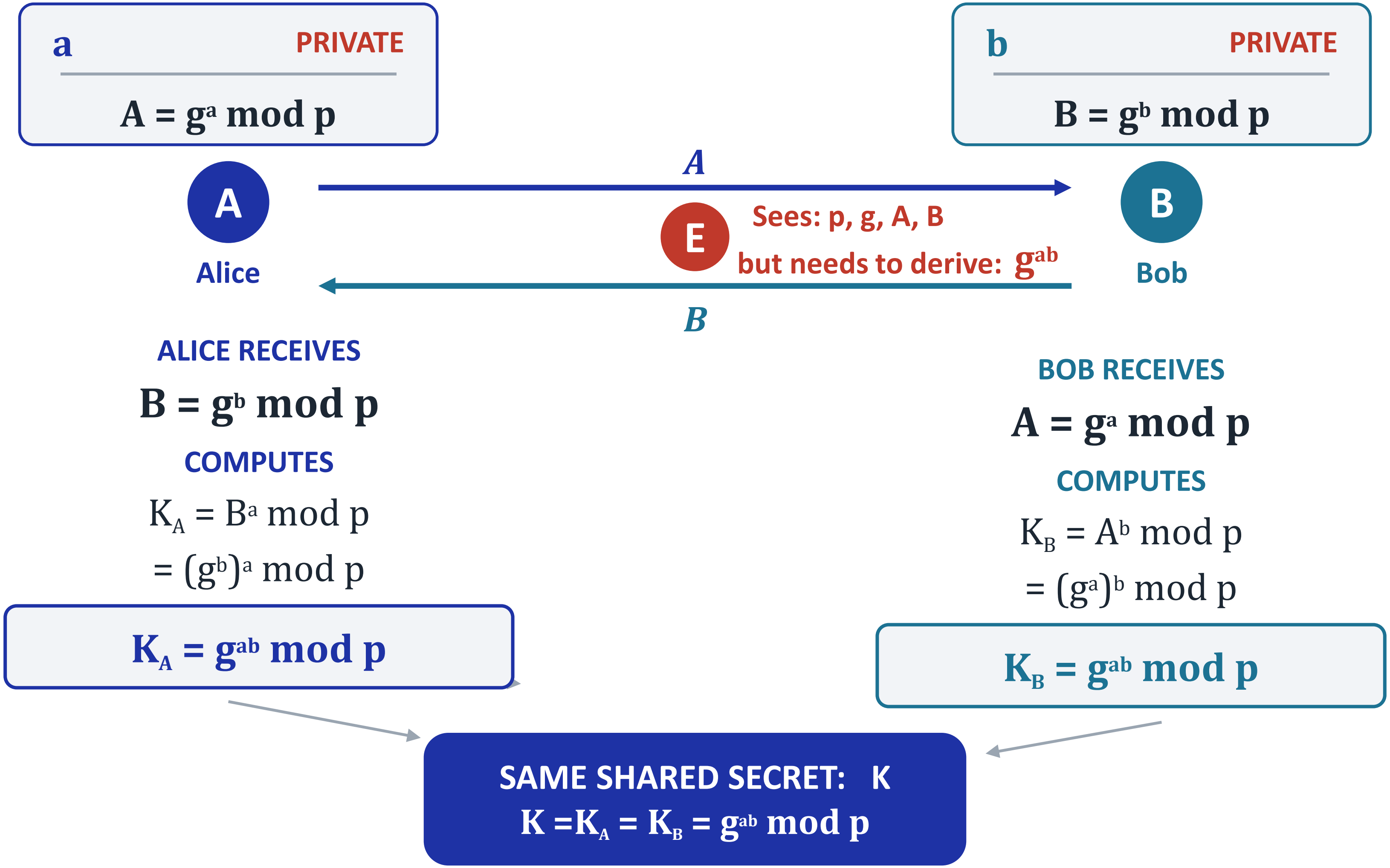
Eve does not know: a, b

Public values are derived from private values using modular exponentiation; the private values are never transmitted.

DIFFIE–HELLMAN PROTOCOL

Alice and Bob Establish the Same Shared Secret

Public parameters: p, g



a PRIVATE

$A = g^{a} \bmod p$

b PRIVATE

$B = g^{b} \bmod p$

Sees: p, g, A, B

but needs to derive: g^{ab}

ALICE RECEIVES

$B = g^{b} \bmod p$

COMPUTES

$K_A = B^{a} \bmod p$

$= (g^{b})^{a} \bmod p$

$K_A = g^{ab} \bmod p$

BOB RECEIVES

$A = g^{a} \bmod p$

COMPUTES

$K_B = A^{b} \bmod p$

$= (g^{a})^{b} \bmod p$

$K_B = g^{ab} \bmod p$

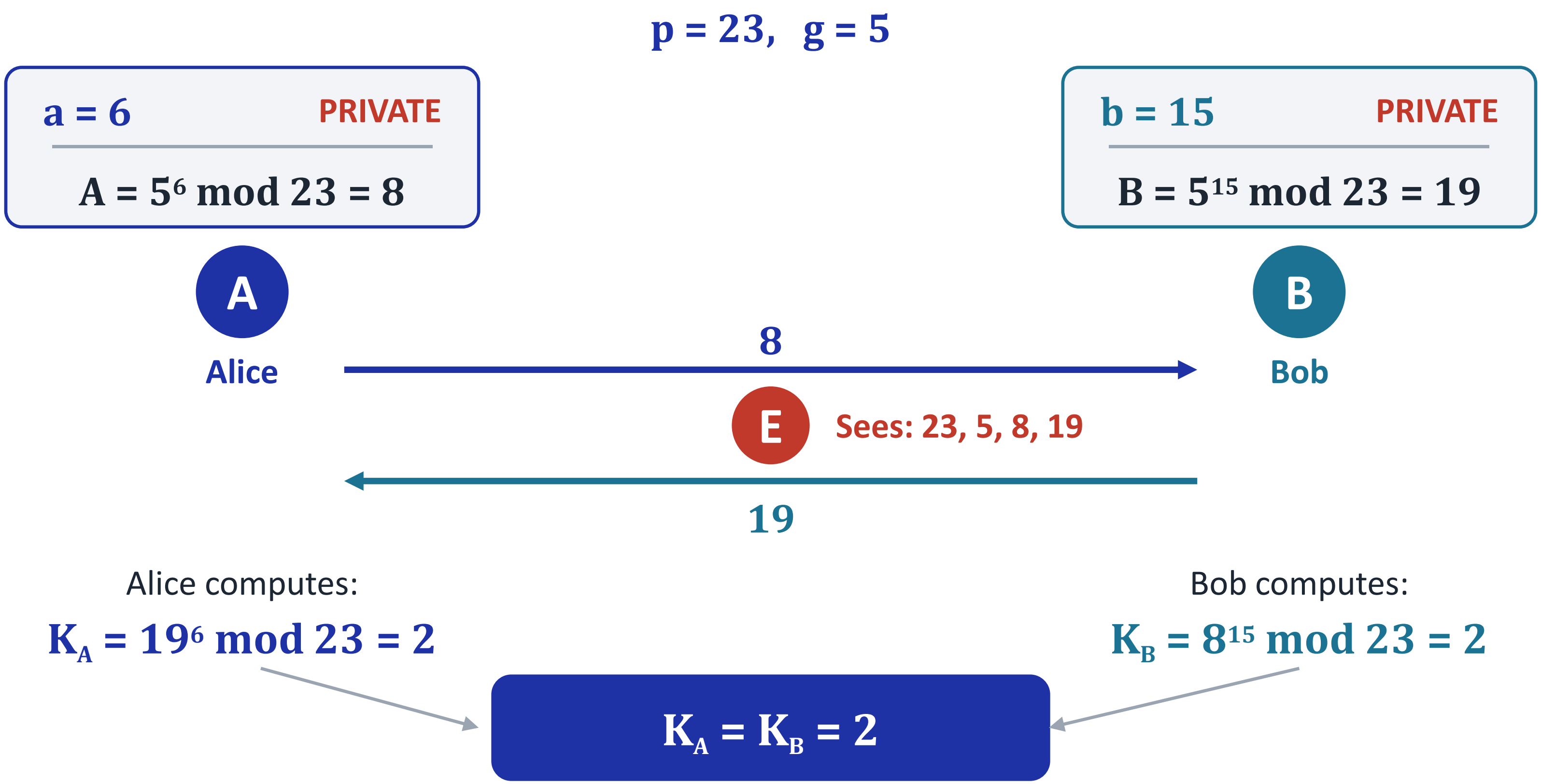
SAME SHARED SECRET: K

$K = K_A = K_B = g^{ab} \bmod p$

Alice and Bob exchange only public values, yet independently compute the same shared secret.

WORKED EXAMPLE

Let's Establish a Shared Secret



$p = 23, \quad g = 5$

$a = 6$ **PRIVATE**

$A = 5^6 \bmod 23 = 8$

A

Alice

$b = 15$ **PRIVATE**

$B = 5^{15} \bmod 23 = 19$

B

Bob

8

E Sees: 23, 5, 8, 19

19

Alice computes:

$K_A = 19^6 \bmod 23 = 2$

Bob computes:

$K_B = 8^{15} \bmod 23 = 2$

$K_A = K_B = 2$

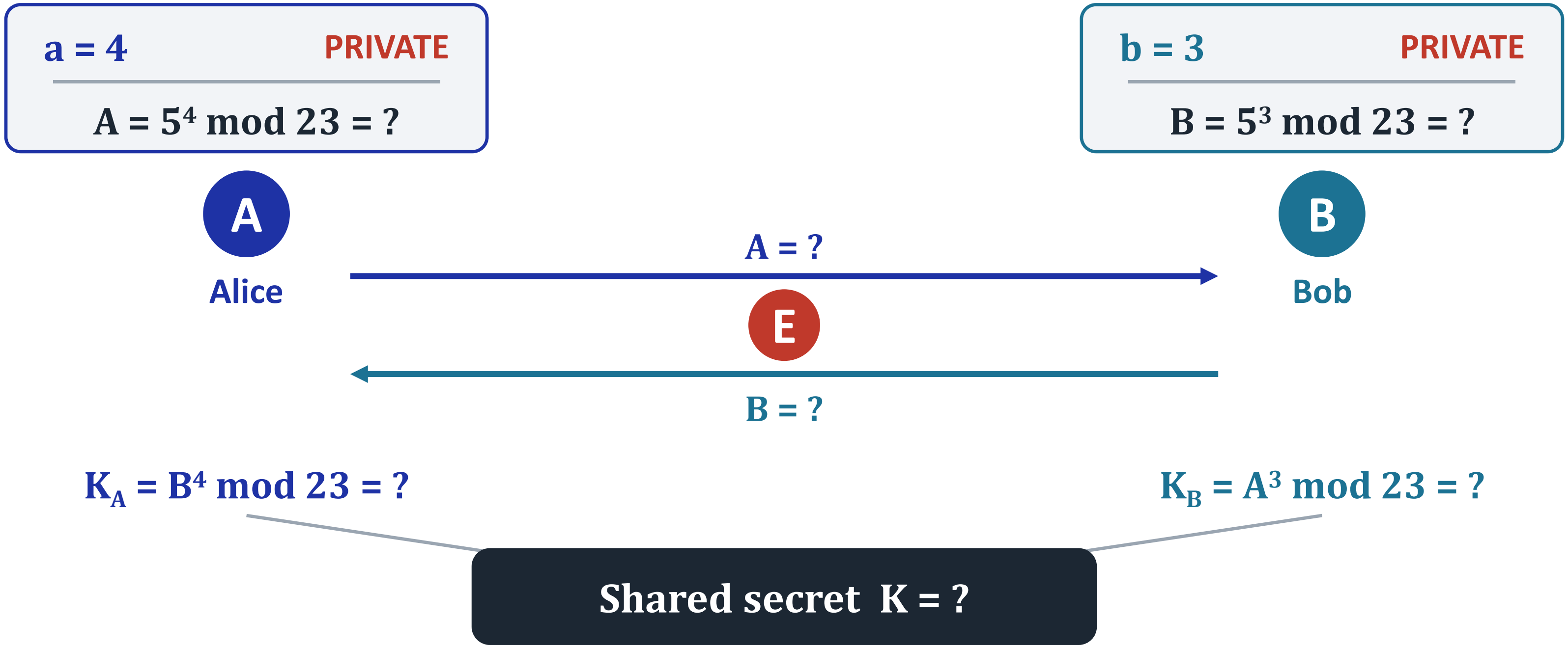
Toy parameters — for learning the computation only.

Different private values + exchanged public values → the same shared secret.

YOUR TURN

Can You Establish the Shared Secret?

Given: $p = 23$, $g = 5$



$a = 4$ **PRIVATE**

$A = 5^4 \bmod 23 = ?$

$b = 3$ **PRIVATE**

$B = 5^3 \bmod 23 = ?$

$A = ?$

$B = ?$

$K_A = B^4 \bmod 23 = ?$

$K_B = A^3 \bmod 23 = ?$

Shared secret $K = ?$

REVEAL

$A = 4,\ B = 10$

$K_A = 18,\ K_B = 18$

$K = 18$

Both sides independently arrive at K = 18.

REAL-WORLD DIFFIE–HELLMAN

Real Systems Use Standardized Groups

Classroom example:
$p = 23, g = 5$

→

Real systems: standardized cryptographic groups

FINITE-FIELD DH

$$A = g^a \bmod p$$

Examples:

`ffdhe2048`

`ffdhe3072`

`MODP Group 14 (2048-bit)`

Applications:

- **TLS**
- **IKEv2**

ELLIPTIC-CURVE DH

Examples:

`X25519`

`secp256r1 (P-256)`

Applications:

- **TLS**
- **SSH**
- **IKEv2**

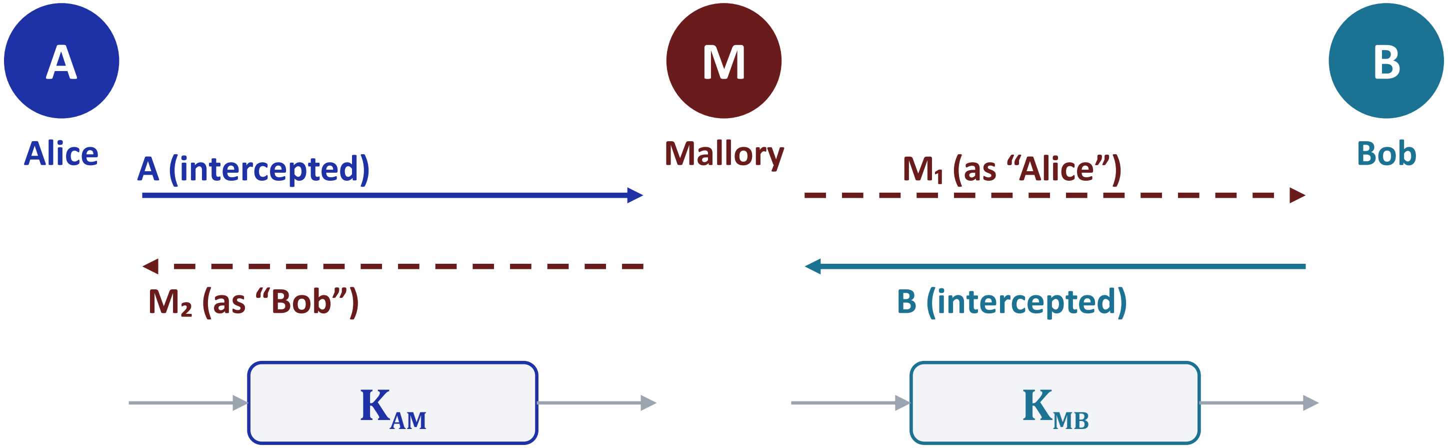
A DH group defines the mathematical setting and parameters used for the key agreement.

Select/agree group → **Ephemeral DH exchange** → **Shared secret** → **KDF** → **Session keys**

Real DH uses carefully selected standardized groups — not classroom-sized parameters.

WHAT COULD GO WRONG?

Diffie–Hellman Agrees on a Secret — But With Whom?



A

Alice

M

Mallory

B

Bob

A (intercepted)

M_1 (as “Alice”)

M_2 (as “Bob”)

B (intercepted)

K_{AM}

K_{MB}

Alice believes K_{AM} is shared with Bob

Bob believes K_{MB} is shared with Alice

There is no Alice–Bob shared secret.

Alice → Encrypt → Mallory **Decrypt → Read/Modify → Re-encrypt** **→ Bob**

Mallory did not solve a discrete logarithm.

Mallory performed two valid DH exchanges.

Missing property: AUTHENTICATION

Basic Diffie–Hellman establishes a shared secret, but does not authenticate the peer.